

Topic 04: Separable Equations & Initial Value Problems (The Clean Split)

Professor Leonard Master Series · Lessons 12 to 14

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Great Divorce

Separable differential equations are the foundation of analytical ODEs. Professor Leonard describes the process as "The Great Divorce":

"You have an equation where x's and y's are mingled together. Your mission is to send all y's to the left side with dy, and all x's to the right side with dx. Once the divorce is finalized, you integrate both sides independently!"

$$\frac{dy}{dx} = g(x)h(y) \implies \frac{1}{h(y)} dy = g(x) dx$$

2. Professor Leonard's Whiteboard Problem Walkthroughs

Problem 4.1: Polynomial Separation & Cubic Roots

Statement: Solve $\frac{dy}{dx} = \frac{x^2}{y^2}$.

Step-by-Step Whiteboard Solution:

- **Step 1: Separate:**

$$y^2 dy = x^2 dx$$

- **Step 2: Integrate both sides:**

$$\int y^2 dy = \int x^2 dx \implies \frac{y^3}{3} = \frac{x^3}{3} + C_1$$

- **Step 3: Clear denominators:** Multiply by 3:

$$y^3 = x^3 + 3C_1 \implies y^3 = x^3 + C$$

- **Step 4: Solve explicitly for y:**

$$y(x) = \sqrt[3]{x^3 + C}$$

Problem 4.2: The "Lost" Singular Solution Trap

Statement: Find the complete general solution of $\frac{dy}{dx} = 6x(y-1)^{2/3}$.

Step-by-Step Whiteboard Solution:

- **Step 1: Note potential division by zero:** To separate, we must divide by $(y-1)^{2/3}$. This assumes $y-1 \neq 0$.

Check singular candidate

: If $y(x) \equiv 1$, then $y' = 0$ and $6x(1-1)^{2/3} = 0$. Thus, $y = 1$ is an **exact equilibrium solution!**

- **Step 2: Separate assuming $y \neq 1$:**

$$(y-1)^{-2/3} dy = 6x dx$$

- **Step 3: Integrate both sides:**

$$\frac{(y-1)^{1/3}}{1/3} = 3x^2 + C_1 \implies 3(y-1)^{1/3} = 3x^2 + C_1$$

- **Step 4: Divide by 3 (relabeling $C = C_1/3$):**

$$(y-1)^{1/3} = x^2 + C$$

- **Step 5: Cube both sides:**

$$y-1 = (x^2 + C)^3 \implies y(x) = (x^2 + C)^3 + 1$$

- **Note on Singular Solution:** Can $(x^2 + C)^3 + 1$ produce $y = 1$? Only if $x^2 + C = 0$ for all x , which is impossible for constant C . Therefore, $y = 1$ is a **singular solution** that cannot be obtained from the general family. Full answer: $y(x) = (x^2 + C)^3 + 1$ and $y = 1$.

Problem 4.3: Initial Value Problem with Natural Logarithms

Statement: Solve the IVP:

$$\frac{dy}{dx} = \frac{2x(y^2 + 1)}{y}, \quad y(0) = 1$$

Step-by-Step Whiteboard Solution:

- **Step 1: Separate variables:**

$$\frac{y}{y^2 + 1} dy = 2x dx$$

- **Step 2: Integrate both sides:**

$$\int \frac{y}{y^2 + 1} dy = \int 2x dx$$

LHS: Let $u = y^2 + 1$, $du = 2ydy \implies ydy = \frac{1}{2}du$.

$$\frac{1}{2} \ln(y^2 + 1) = x^2 + C_1$$

(No absolute value needed because $y^2 + 1 > 0$ for all real y)

- **Step 3: Clear the fraction and exponentiate:**

$$\ln(y^2 + 1) = 2x^2 + 2C_1$$

$$y^2 + 1 = e^{2x^2 + 2C_1} = e^{2C_1} \cdot e^{2x^2} = Ae^{2x^2} \quad (A > 0)$$

$$y^2 = Ae^{2x^2} - 1$$

- **Step 4: Apply $y(0) = 1$:**

$$1^2 = Ae^0 - 1 \implies 1 = A - 1 \implies A = 2$$

$$y^2 = 2e^{2x^2} - 1 \implies y(x) = \pm \sqrt{2e^{2x^2} - 1}$$

- **Step 5: Select the correct branch:** Because $y(0) = +1 > 0$, choose the positive square root:

$$y(x) = \sqrt{2e^{2x^2} - 1}$$

Problem 4.4: Trigonometric Separation & Domain Blow-Up

Statement: Solve $\frac{dy}{dx} = y^2 \sin x$, $y(0) = 1$, and find the interval of definition.

Step-by-Step Whiteboard Solution:

- **Step 1: Separate:**

$$y^{-2} dy = \sin x dx$$

- **Step 2: Integrate:**

$$-y^{-1} = -\cos x + C \implies \frac{1}{y} = \cos x - C$$

- **Step 3: Apply $y(0) = 1$:**

$$\frac{1}{1} = \cos(0) - C \implies 1 = 1 - C \implies C = 0$$

- **Step 4: Invert:**

$$\frac{1}{y} = \cos x \implies y(x) = \frac{1}{\cos x} = \sec x$$

- **Step 5: Interval of definition:** The function $\sec x$ blows up to ∞ at $x = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots$. The initial condition is given at $x_0 = 0$. The largest continuous open interval containing 0 is:

$$I = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Problem 4.5: Integration by Substitution on the RHS

Statement: Solve the IVP:

$$\frac{dy}{dx} = \frac{xy^3}{\sqrt{1+x^2}}, \quad y(0) = -1$$

Step-by-Step Whiteboard Solution:

- **Step 1: Separate:**

$$y^{-3}dy = \frac{x}{\sqrt{1+x^2}}dx$$

- **Step 2: Integrate both sides:**

$$\int y^{-3}dy = \int x(1+x^2)^{-1/2}dx$$

LHS: $\frac{y^{-2}}{-2} = -\frac{1}{2y^2}$. RHS: Let $u = 1 + x^2$, $du = 2xdx$. $\int \frac{1}{2}u^{-1/2}du = u^{1/2} + C = \sqrt{1+x^2} + C$.

$$-\frac{1}{2y^2} = \sqrt{1+x^2} + C$$

- **Step 3: Apply $y(0) = -1$:**

$$-\frac{1}{2(-1)^2} = \sqrt{1+0} + C \implies -\frac{1}{2} = 1 + C \implies C = -\frac{3}{2}$$

- **Step 4: Solve for $y(x)$:**

$$-\frac{1}{2y^2} = \sqrt{1+x^2} - \frac{3}{2} \implies \frac{1}{2y^2} = \frac{3 - 2\sqrt{1+x^2}}{2}$$

$$y^2 = \frac{1}{3 - 2\sqrt{1+x^2}}$$

Since $y(0) = -1 < 0$, choose the **negative square root**:

$$y(x) = -\frac{1}{\sqrt{3 - 2\sqrt{1+x^2}}}$$

Problem 4.6: Implicit Factoring & Integration by Parts

Statement: Solve $e^x y \frac{dy}{dx} = e^{-y} + e^{-2x-y}$.

Step-by-Step Whiteboard Solution:

- **Step 1: Factor the RHS:**

$$e^{-y} + e^{-2x-y} = e^{-y}(1 + e^{-2x})$$

- **Step 2: Separate x and y :**

$$e^x y \frac{dy}{dx} = e^{-y}(1 + e^{-2x}) \implies \frac{y}{e^{-y}} dy = \frac{1 + e^{-2x}}{e^x} dx$$

$$ye^y dy = (e^{-x} + e^{-3x}) dx$$

- **Step 3: Integrate LHS by parts:** Let $u = y, dv = e^y dy \implies du = dy, v = e^y$.

$$\int ye^y dy = ye^y - \int e^y dy = (y - 1)e^y$$

- **Step 4: Integrate RHS:**

$$\int (e^{-x} + e^{-3x}) dx = -e^{-x} - \frac{1}{3}e^{-3x} + C$$

- **Step 5: Full implicit general solution:**

$$(y - 1)e^y + e^{-x} + \frac{1}{3}e^{-3x} = C$$

3. Common Exam Traps & Professor Leonard Warnings

- **Trap 1: Dropping $+C$ inside algebra:** Writing $\ln y = x \implies y = e^x + C$ instead of $y = e^{x+C} = Ae^x$.
- **Trap 2: Ignoring Square Root Signs:** When taking $y = \pm\sqrt{\dots}$, you must check the initial condition $y(x_0) = y_0$ to choose the correct $+/ -$ sign.
- **Trap 3: Forgetting Singular Solutions:** Always check if dividing by $h(y)$ throws away an equilibrium solution $h(y) = 0$.